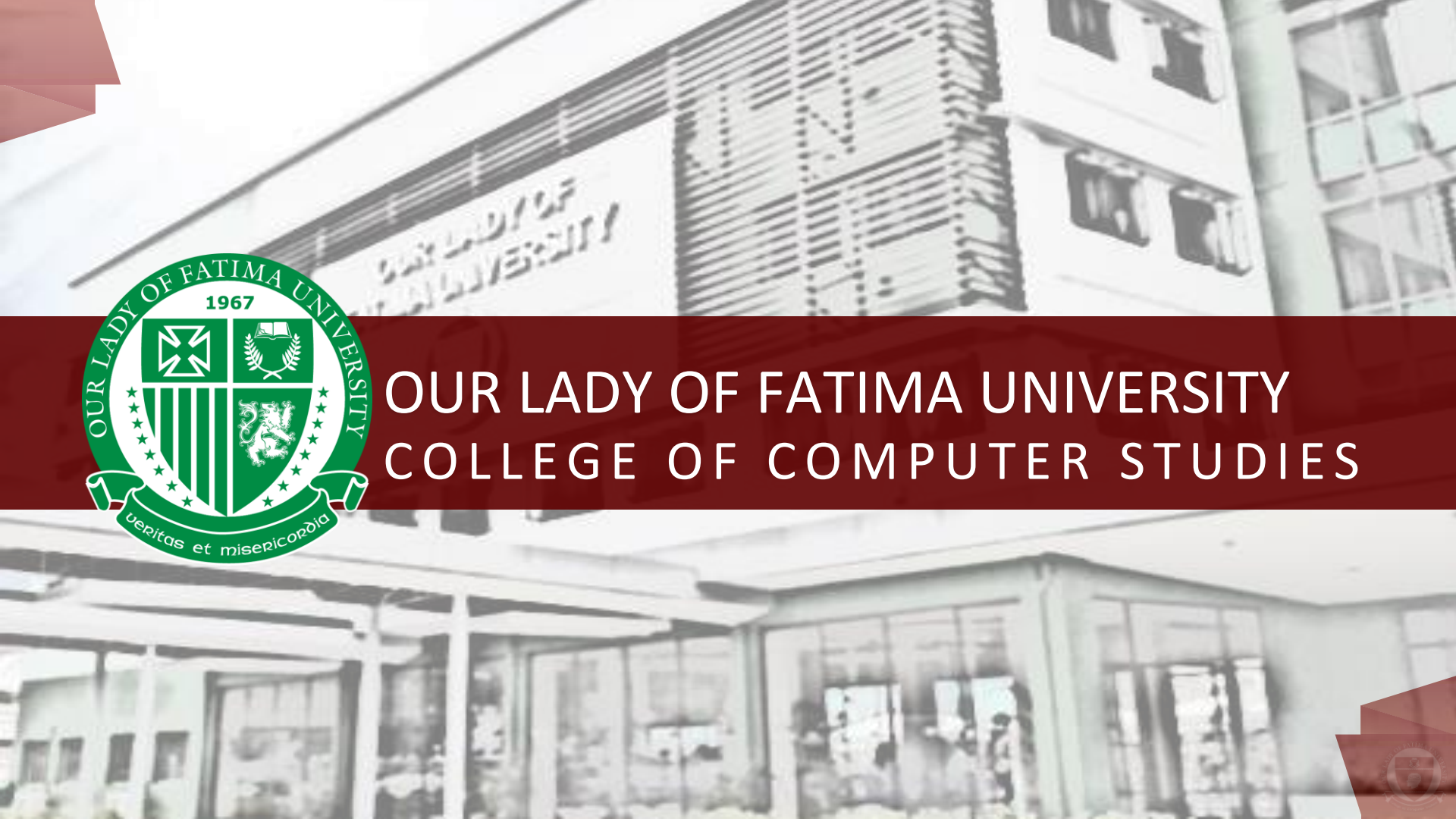
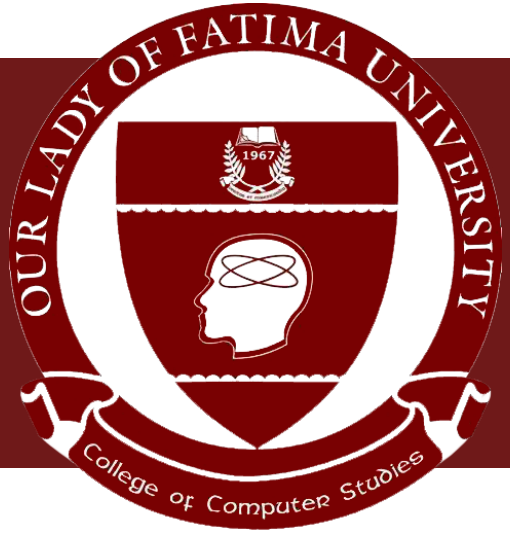
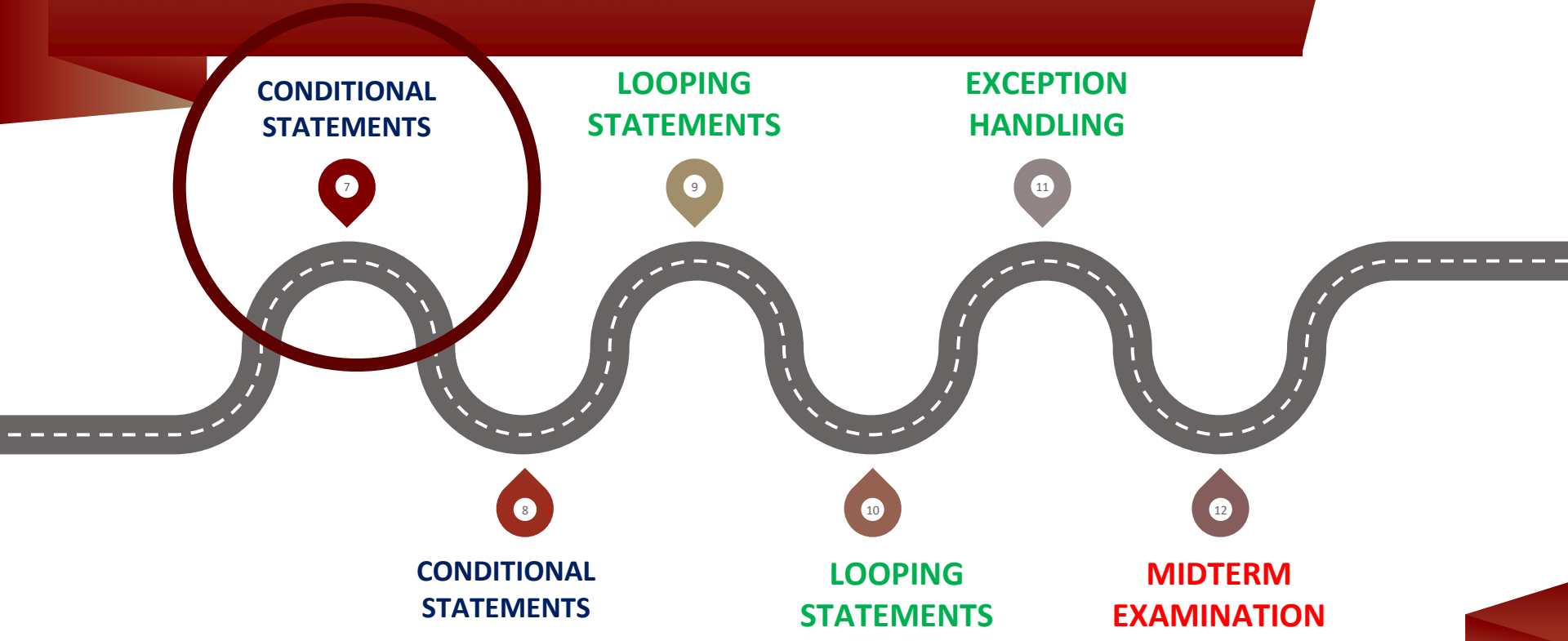


OUR LADY OF FATIMA UNIVERSITY
COLLEGE OF COMPUTER STUDIES



OOPR211
OBJECT-ORIENTED
PROGRAMMING I

MIDTERM ROADMAP



Module 5: JAVA CONTROL STATEMENTS – CONDITIONAL STATEMENTS

Module Objectives:

1

Define what is a control structure and differentiate the types of conditional statements.

2

Practice problem solving solution using a different approach like conditionals statement.

Topics:

- Understanding if statements
- Complex conditionals
- The switch, case, and break statements



Java Statement

In Java, a *statement* is one or more lines of code ending with a semi-colon (;).

It generally contains expressions (expressions have a value).

Java Statement

- **Simple Statement**
  - Basic building blocks of a program
  - One or more lines terminated by a semicolon (;)

```
printf(Hello, World!”);
```

```
int main() {
   System.out.println (“Hello, ”);
   System.out.println (“World!”);
}
```

- **Compound Statement or Block**
  - Used to organize simple statements into complex structures
  - One or more statements enclosed by braces { }
  - Recommended to indent blocks for readability



Control Structure

- **Control structures allow to change the ordering of how the statements in a program is executed**
- Types of control structures:
  - ***Decision control structures***
    - allow to select specific sections of code to be executed
  - ***Repetition control structures***
    - allow to execute specific sections of the code a number of times

Control Structure

- Java statements that allows a programmer to select and execute specific blocks of code while skipping other sections.
- Include types such as:
  - **`If`**
  - **`If-Else`**
  - **`If-Else-If`**
  - **`Switch structure`**

1

Understanding *if* statements

THE *if* STATEMENT

The simple if statement has the following syntax:

if (<boolean expression>)

<statement action>

THE IF STATEMENT EXECUTES A BLOCK OF CODE ONLY IF THE SPECIFIED EXPRESSION IS TRUE.

THE *if* STATEMENT

```
public class IfStatementDemo
{

        public static void main(String[] args)
        {

                int a = 10, b = 20;
                if (a > b)
                        System.out.println("a > b");
                if (a < b)
                        System.out.println("b > a");
        }
}
```

Output:

b > a

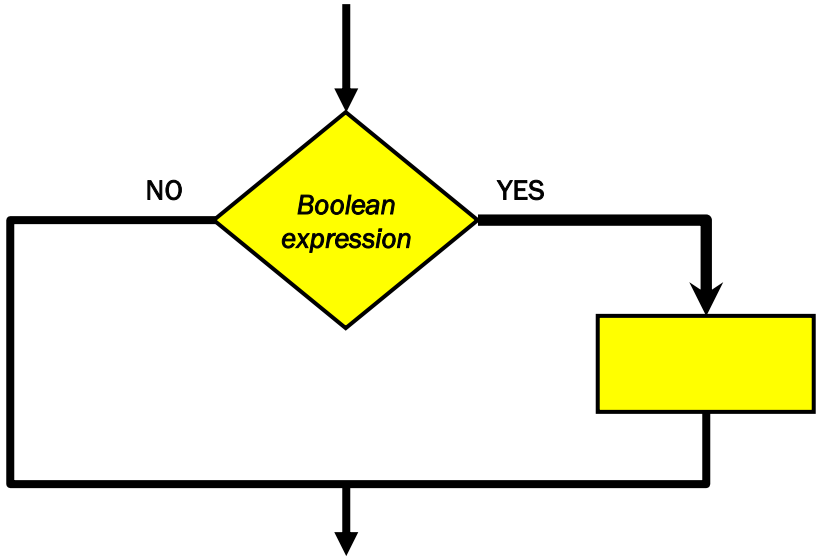
THE *if* STATEMENT

Every time Java encounters an *if* statement,

1. It determines whether *expression* is true or false.
2. If *expression* is true, then Java executes *statement*.
3. If *expression* is false, then Java ignores or skips the remainder of the *if* statement and proceeds to the next statement.

THE *if* STATEMENT

Making a decision involves choosing between two alternative courses of action based on some value within a program.



THE *if* STATEMENT

If Structure: Example 1

- Suppose that the passing grade on an examination is 75 (out of 100). Then the if statement may be written in Java as:

```
if (grade >= 75)
    System.out.println("You Passed!");
```

THE *if* STATEMENT

If Structure: Example 2

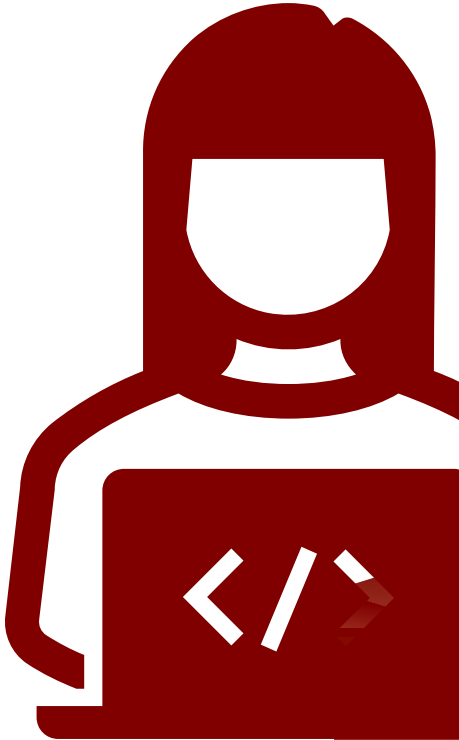
```
int grade = 68;
if (grade > 60)
{
   System.out.println("Congratulations!");
   System.out.println("You Passed!");
}
```

EXERCISES:

1. Test if the value of the variable count is greater than 10. If the answer is yes, print “Count is greater than 10”.

2. Test if the value of the variable number is an odd number. If the answer is yes, print “This number is not an even number.”

3. Test if the value of the variable age is greater than 18 and less than or equal to 40. If the answer is yes, print “You’re not getting old!”



2 Complex conditionals

THE *if-else* STATEMENT

The simple if-else statement has the following syntax:

```
if (<boolean expression>)
        <statement 1>
else
        <statement 2>
```

THE IF/ELSE STATEMENT IS AN EXTENSION OF THE IF STATEMENT. IF THE STATEMENTS IN THE IF STATEMENT FAILS, THE STATEMENTS IN THE ELSE BLOCK ARE EXECUTED.

THE *if-else* STATEMENT

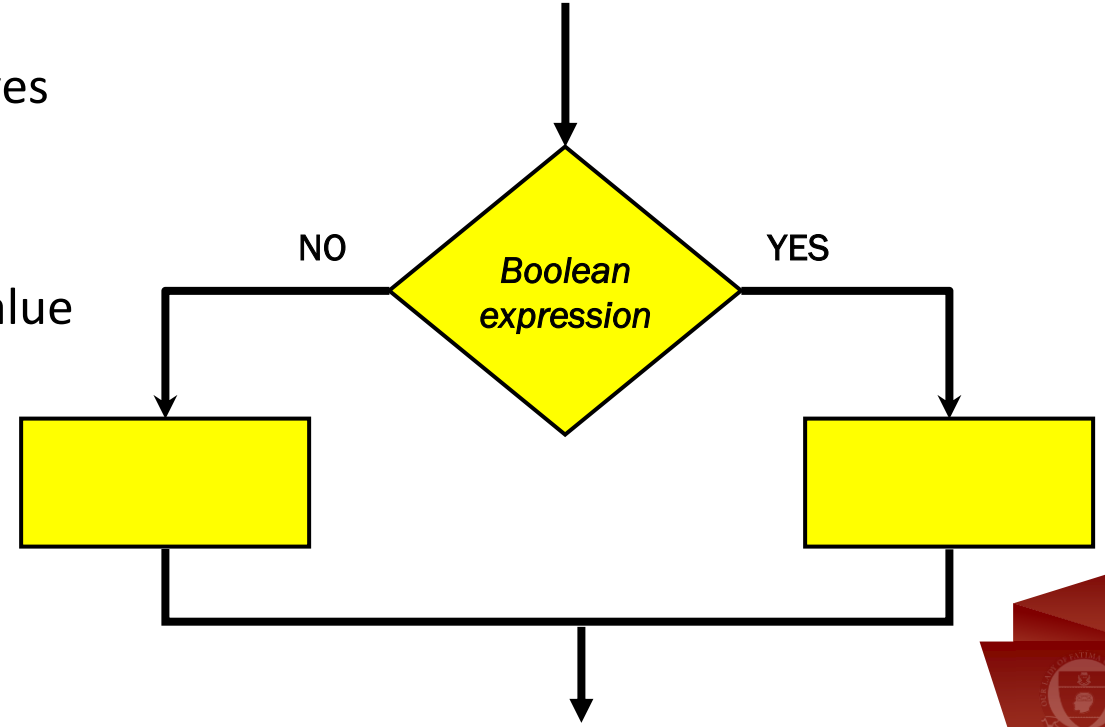
```
public static void main(String[ ] args)
{
        int a = 10, b = 20;
        if (a > b)
        {
                System.out.println("a > b");
        }
        else
        {
                System.out.println("b > a");
        }
}
}
```

Output:

b > a

THE *if-else* STATEMENT

- Making a decision involves choosing between two alternative courses of action based on some value within a program.



THE *if-else* STATEMENT

If-Else Structure: Example 1

```
int grade = 68;
if (grade > 60)
   System.out.println("Congratulations!");
else
 System.out.println("Sorry you failed!");
```

THE *if-else* STATEMENT

If-Else Structure: Example 2

```
int grade = 68;
if (grade > 60) {
 System.out.println("Congratulations!");
   System.out.println("You Passed!");
}
else {
    System.out.println("Sorry you failed!");
}
```

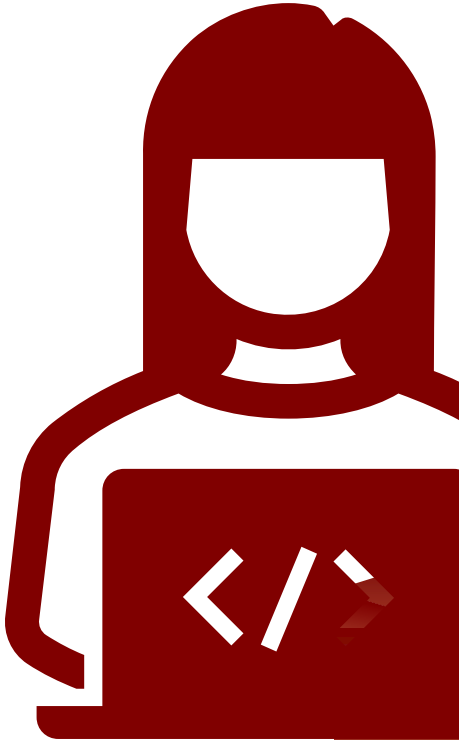
THE *if-else* STATEMENT

If-Else Structure: Example 3

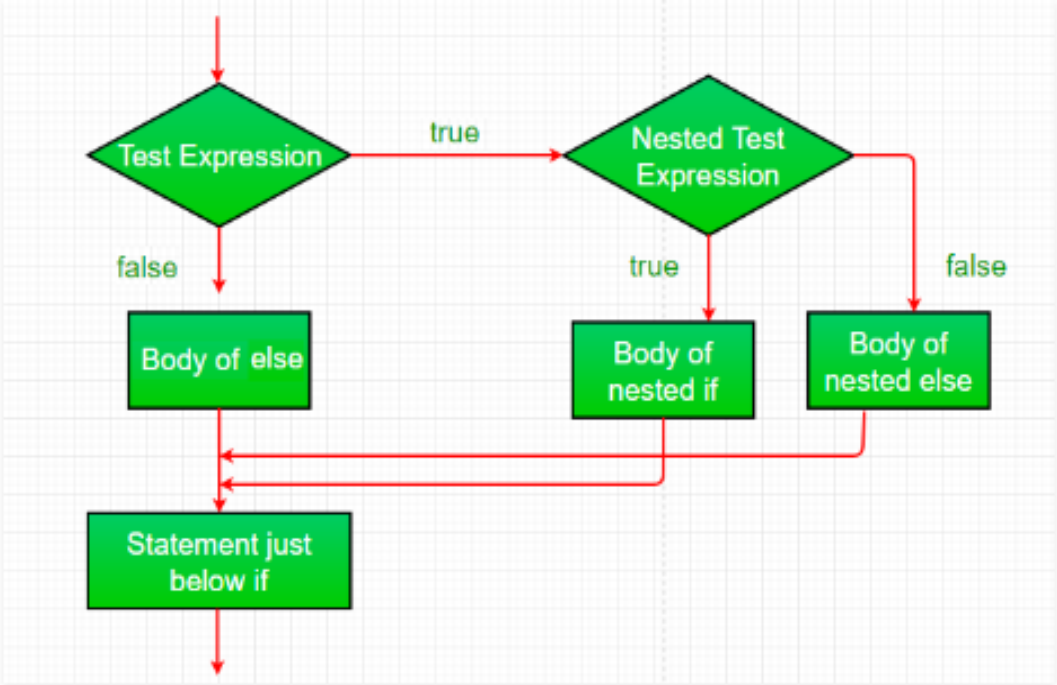
```
int choice = 0;
if (choice == 1)
{
   num = 1;
   num2 = 10;
}
else
{
   num = 2;
   num2 = 20;
}
```

EXERCISES:

1. Test if the value of the variable count is greater than 10. If it is, print “Count is greater than 10”, otherwise, print “Count is less than 10”.
2. Test if the value of the variable number is an odd number. If it is, print “This number is not an even number” otherwise, print “This number is an odd number.”
3. Test if the value of the variable age is greater than 18 and less than or equal to 40. If it is, print “You’re not getting old!” otherwise, print “You’re not getting younger.”



NESTED *if* STATEMENT



NESTED *if* STATEMENT

- A nested if is an if statement that is the target of another if or else. Nested if statements means an if statement inside an if statement. Yes, java allows us to nest if statements within if statements. i.e, we can place an if statement inside another if statement.

```
if (condition1)
{
 // Executes when
condition1 is true
 if (condition2)
      {
 // Executes when
condition2 is true
     }
 }
```

3

The switch, case, and break statements

switch case STATEMENT

ALSO CALLED A CASE STATEMENT IS A MULTI-WAY BRANCH WITH SEVERAL CHOICES.

A SWITCH IS EASIER TO IMPLEMENT THAN A SERIES OF IF/ELSE STATEMENTS.

switch case **STATEMENT**

```
switch(Controlling_Expression)
{
        case Case_Label:
        statement;
        statement;
        ……
        statement;
        break;

        case Case_Label:
        statement;
        statement;
        ……
        statement;
        break;
 default:
        statement;
        statement;
        ……
        statement;
        break;
}
```

switch case STATEMENT

break statement

- defines the end of a case.

switch case STATEMENT

default statement

- use in defining message to be displayed that is outside of the given case.

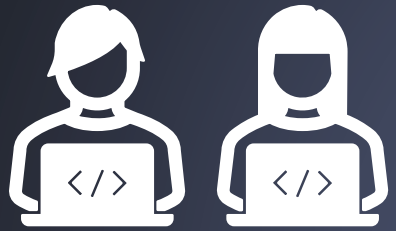
```
public class SwitchCaseStatementDemo
{
        public static void main(String[] args)
        {
                int a = 10, b = 20, c = 30;
                int status = -1;
                if (a > b && a > c) {
                        status = 1;
                } else if (b > c) {
                        status = 2;
                } else {
                        status = 3;
                }
                switch (status) {
                case 1:
                        System.out.println("a is the greatest");
                        break;
                case 2:
                        System.out.println("b is the greatest");
                        break;
                case 3:
                        System.out.println("c is the greatest");
                        break;
                default:
                        System.out.println("Cannot be determined");
                }
        }
}
```

Output

c is the greatest

```
int grade = 92;

switch (grade) {
case 100:
System.out.println("Excellent!");
break;
case 90:
System.out.println("Good!");
break;
case 80:
System.out.println("Study Harder!");
break;
default:
System.out.println("You failed!");
}
```



Check your Understanding

Write a program to output summation of a set of numbers.

Example Output:

Please input how many numbers you would like to add: 4

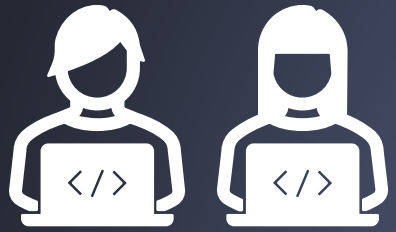
Please enter a value: 3

Please enter a value: 1

Please enter a value: 2

Please enter a value: 3

The sum of the 4 values is 9



Check your Understanding

Write a program which should be able to reproduce the following example output:

```
*
***
*****
*******
*********
```

QUESTIONS?

Thank you!

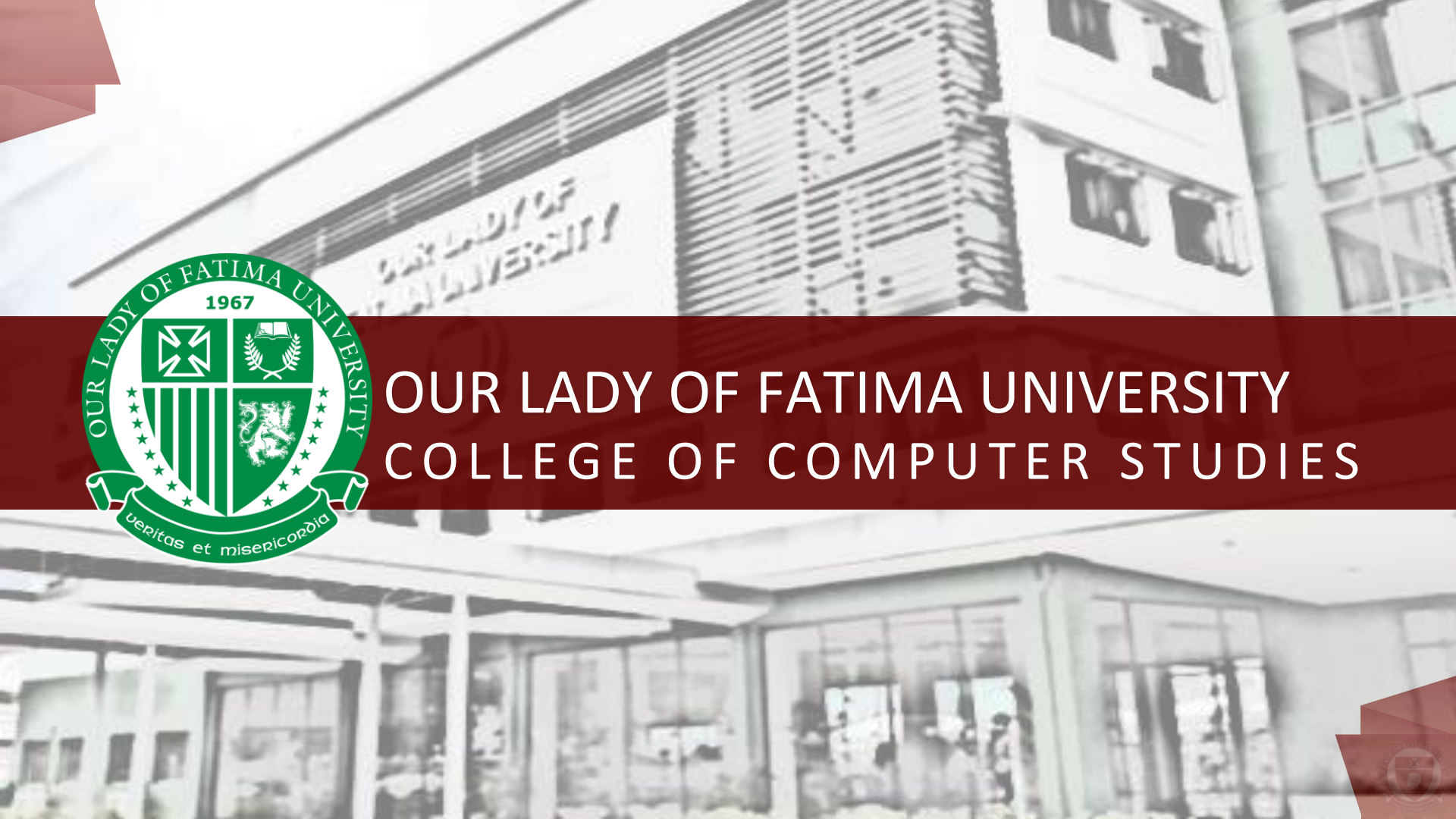
Credits

Special thanks to all the people who made and released these awesome resources for free:

- Presentation template by SlidesCarnival
- Photographs by Unsplash

Additional References:

- https://www.edureka.co/blog/what-is-java/
- http://ecomputernotes.com/java/what-is-java-language/types-of-java-programs
- https://app.codechum.com/teacher/classes/504/timeline/lessons/java-conditionals/if-statement



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